

ISM 6251 and 4543: Machine Learning (Fall 2025)

Assignment 1

- Please provide succinct answers to the questions below.
- You should submit a .ipynb file and a pdf (for any non-coding questions) in Canvas.
- Please include the names of all team-members in your write up and in the name of the file. One submission per group.
- You should include all the Python functions you use.

Problem 1. Download the chickwts.csv from Canvas and do the following tasks.

- (a) How many instances and variables are in this data set? How do you get this information without viewing data?
- (b) The column “weight” means weights in grams. Add a “new” column that gives the weight in kilograms, rather than grams. Make sure the column has a descriptive name.
- (c) Make a box-and-whiskers plot showing the distribution of chicken weight according to feed type. Make sure to label the axes appropriately.

Problem 2. Download the OVCclinicalinfo.csv from Canvas. Import the spreadsheet and name it as a dataframe “clin”. Briefly inspect the data.

- (a) Summarize the distribution of each variable in this data set.
- (b) Remove the variable “Survival” from this data set. Convert the first column to character type. Change the name of the third column to “event”. Convert the “CA125.POST” and “GRADE” columns into numeric values, with the ambiguous entries coded as NAs. Fix the “Debulk” column such that only two levels are used (“O” and “S”). Rename the eighth column to “response” and convert it to a binary variable (0 = FALSE, 1 = TRUE)
- (c) Is there any missing values or outliers in this data set? If yes, how do you handle them? Justify your choice.

Problem 3. Download the “hmeq.csv” data from Canvas. Write down a Python code that perform data cleaning such as dealing with missing values, and removes all the outliers with respect to all the numerical variables “together”.

Hint: You can write your own function (lets call it RemoveOutlier; see below) that receives the variable x as its input and returns the values of x that are not outlier. Be creative.

```
def RemoveOutlier(x)
    Your code
    return( object that you need to use outside of the function )
```

Problem 4. Use the “hmeq.csv” to construct a decision tree model.

1. Before modeling, split the data into two subsets: training (70%) and test (30%).
2. Ensure that the target variable BAD is a flag variable.
3. Then build the decision tree and check the prediction result using the default settings, and take a look at the tree by printing the rules and plotting the tree. Explain and discuss the rules and plot briefly.
4. Manipulate the default setting by changing the parameters, and try to improve the performance. Justify your choice.
5. Once you nail down the model parameters, get the feature importance.

Problem 5. Please do not use Python to answer this question. Consider the following data set that contains some hypothetical information about 10 USF students who have taken the machine learning class. We want to construct a decision tree to predict who passes the final exam.

Instances	Department	Master?	Number of courses taken	Pass?
1	ISM	Y	3	Y
2	ISM	Y	7	N
3	ISM	N	4	Y
4	IE	N	4	N
5	IE	Y	7	N
6	ISM	Y	4	Y
7	IE	N	5	Y
8	ISM	N	6	Y
9	IE	Y	1	Y
10	ISM	Y	5	N

- Given the 10 instances in the table above, calculate the information gain of each candidate split point. Which attribute do you choose at the root of your tree?
- Construct the entire tree to its full depth. Show your computations and the final tree.
- Using your decision tree, provide two decision rules that the instructor can use to predict if a student is going to pass or fail. Justify your choice.
- If we use Gini index to construct a decision tree, which variable will be the root node? Explain.